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To cite this article: Han Li , Liang Zhang , Huiwen Zhou & Tianzhen Hong (05 Apr 2026): MCP-enabled agentic AI workflow for building energy modelling: framework and use cases, Journal of Building Performance Simulation, DOI: [10.1080/19401493.2026.2653969](https://doi.org/10.1080/19401493.2026.2653969)

To link to this article: <https://doi.org/10.1080/19401493.2026.2653969>



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Published online: 05 Apr 2026.



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MCP-enabled agentic AI workflow for building energy modelling: framework and use cases

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ABSTRACT

Traditional building energy modelling workflows remain labor-intensive and error-prone, requiring specialized expertise that limits broader adoption. This paper introduces a novel Model Context Protocol (MCP)-enabled framework that connects AI assistants to EnergyPlus through MCP, a standardized interface for tool invocation and context management. Two complementary integration paradigms are presented and compared: conversational integration, where users interact through natural language while an AI assistant orchestrates MCP tools on demand, and agentic workflow integration, where specialized agents coordinate autonomously to complete multi-step tasks. Using an experimental testbed for residential buildings, the end-to-end workflows are demonstrated. The conversational approach reduced typical inspection and modification tasks from 1–2 h to under 15 min, while maintaining full transparency through visible tool invocations. The agentic approach automated parametric analysis. These demonstrations establish MCP as a foundational layer for AI-assisted building energy modelling, enabling natural language interactions with simulation tools while preserving professional oversight and decision-making authority.

Highlights

- An MCP-enabled framework provides standardized integration between AI assistants and EnergyPlus for building energy modelling workflows.
- Two complementary paradigms are demonstrated: conversational integration for exploratory tasks and agentic workflow integration for systematic automation.
- End-to-end demonstrations on a residential energy model show an 80% to 90% time reduction for model inspection, modification, and analysis tasks.
- The framework augments rather than replaces professional judgment, with AI handling tool orchestration while practitioners retain decision-making authority.
- The approach establishes a foundational infrastructure for advanced capabilities including parametric simulations, BIM-to-BEM translation, and automated model calibration.

ARTICLE HISTORY

Received 5 January 2026
Accepted 26 March 2026

KEYWORDS

Agentic AI; building energy modeling (BEM); large language model (LLM); Model Context Protocol (MCP); energy and buildings

1. Introduction

1.1. Background

Building energy modeling (BEM), also referred to as building energy performance modelling and/or simulation, started around the 1970s to model the dynamic thermal, energy, and environmental performance of buildings and energy systems (Hong, Chou, and Bong 2000). BEM has achieved accelerating adoption among researchers and practitioners to inform critical decision-making on energy efficiency, demand flexibility, and indoor environmental quality (IEQ) for both the design of new buildings and the retrofit, operations and controls of existing buildings.

Historically, BEM relied on traditional physics-based modelling. However, the field has significantly expanded to include reduced-order modelling. More recently, BEM has integrated data-driven modelling, leveraging the power of machine learning and deep learning. This evolution means that modellers now have the flexibility to choose one approach or hybrid modelling methods. This selection is based on a fit-for-purpose framework, allowing them to balance the level of required resources, necessary detail, and accuracy for specific use cases.

Despite significant advancements and increasing adoption within the industry, BEM workflows face several persistent challenges (Hong, Langevin, and Sun 2018).

One major area of difficulty is the automation and improvement of the BEM simulation workflow. This includes the labor-intensive process of collecting and preparing input data required for creating complex energy models. Additional challenges in the workflow involve the automated generation of models, the complexities of debugging and calibrating models, and the effort needed to interpret the large volume of simulation results to extract actionable insights for decision-making. BEM also faces the challenge of multi-scale integration. This involves coupling building energy models with other large-scale system models – such as transportation, power grids, and urban microclimate models at the regional or urban level – in order to understand and improve performance of integrated urban energy and environmental systems.

The recent confluence of affordable, powerful computing, and an increasing volume of data collected from buildings and systems has positioned artificial intelligence (AI) to transform BEM workflows across a building's entire life cycle, informing decision making for occupant comfort, energy efficiency, demand flexibility, and energy resilience. AI offers unprecedented opportunities to address long-standing challenges in BEM related to automation, fidelity, and multi-scale integration, driven by powerful computing and advancements in data infrastructure.

AI can transform the BEM ecosystem across data, models, computing, and applications (Hong and Zhang 2025). In the Data theme, AI techniques like computer vision and natural language processing automate building characterization and inventory creation, reducing the need for costly manual site audits. AI tools analyze sources such as Lidar scans, satellite images, and design drawings to extract critical BEM inputs, including building geometry, thermal zoning, construction type, and inferred HVAC system details. AI also cleans, completes, and transforms diverse input data, filling gaps and inferring missing details to create AI-ready and model-ready data. For Models, AI provides alternatives to and enhancements for physics-based BEM, e.g. surrogate models and deep learning models accelerate complex and large-scale simulations.

1.2. Recent advancements in generative AI for BEM

1.2.1. Overview and key AI technologies

Recent advances in generative AI (GenAI), particularly the rapid progress of large language models (LLMs), are reshaping approaches to automation and decision-making in complex engineering domains (Vaithilingam, Zhang, and Glassman 2022; Wang and Chen 2023; Zhang

et al. 2023). Beyond simple natural language processing tasks, LLMs now support knowledge extraction, code generation, simulation, and reasoning for decision-making, making them powerful enablers of intelligent automation in the building energy sector. Their versatility reduces the reliance on specialized expertise for traditionally labor-intensive tasks, thereby establishing a scalable paradigm for system design and digital transformation (Li et al. 2023).

Definition and key concepts

GenAI refers to models capable of both generating and reasoning over content by leveraging large-scale training data, with LLMs representing the most prominent example. Unlike traditional automation tools, GenAI not only processes structured inputs but also extracts semantic meaning from unstructured natural language and generates executable task descriptions. As a result, complex problem-solving no longer depends solely on rule-based scripts but instead benefits from the model's semantic understanding and contextual adaptability (Hou et al. 2026).

The notion of agentic workflows further extends the application scope of GenAI. Rather than relying on a single 'agent', these workflows emphasize the coordinated interaction of multiple agents, tools, and protocols. At their core lies a plan – act – check cycle: setting a goal, invoking tools for execution, and iteratively verifying outcomes. Compared with traditional macros or scripts developed for specific workflows, agentic workflows are adaptive and autonomous, making them well suited to managing complex and dynamic task flows.

Consequently, the integration of GenAI with agentic workflows marks a paradigm shift from 'single-point intelligence' to 'system-level intelligence'. In domains such as BEM, this integration not only lowers the threshold for model development but also enables natural interaction and closed-loop control among humans, models, and simulation engines through contextual protocols such as the Model Context Protocol (MCP) (Anthropic 2024), a standardized interface that allows AI assistants to invoke external tools, query data sources, and interact with simulation engines in a unified way (detailed in Section 1.3).

Automation in domain-specific applications

The integration of automation into domain-specific applications is increasingly driven by LLM agents and agentic workflows. Unlike conventional rule-based approaches, these agents combine natural language reasoning with tool execution, memory, and self-reflection, thereby enabling more adaptive and scalable solutions across both building and non-building domains.

Across domains such as healthcare, education, finance, and software engineering, LLM agents have demonstrated their capacity to perform tasks requiring advan

ced reasoning and domain-specific automation. In healthcare, multi-agent systems such as MedAide (Yang et al. 2025) and MDAgents (Kim et al. 2024) coordinate diagnosis, prescription validation (Van et al. 2024), and treatment planning, enhancing accuracy under uncertainty (Yu et al. 2024). In education, embodied (Zheng et al. 2025) and multi-agent teaching frameworks (Xie and Xu 2025) simulate classroom dynamics, providing personalized learning experiences and real-time feedback (Zhang et al. 2024). In finance, agentic frameworks like FinCon and specialized trading agents automate risk control (Yu et al. 2024), compliance checking (Yu et al. 2024), and collaborative decision-making across multiple analyst roles (Okpala, Golgoon, and Kannan 2025). Within software engineering, multi-agent workflows are applied in requirements engineering (He, Treude, and Lo 2025), accessibility testing (Xia et al. 2024), compiler fuzzing (Yang et al. 2025), automated code review (Rasheed et al. 2024), and even robotics planning (Moncada-Ramirez et al. 2025). Collectively, these cross-domain advancements illustrate how automation has shifted from reliance on static scripts to autonomous, adaptive workflows that integrate reasoning, action, and verification.

In the building sector, automation enabled by LLM agents is still at an early but promising stage. Initial explorations have shown that GenAI-based tools can support model generation, simulation setup, and semantic metadata construction. These studies demonstrate that GenAI is beginning to penetrate the building modelling workflow, bridging natural language descriptions and executable simulation tasks. While such examples remain preliminary, they highlight the feasibility of applying multi-agent reasoning and tool orchestration to building energy modelling – a topic further detailed in Section 1.2.3, which focuses specifically on existing agentic workflows in the BEM domain.

Beyond modelling, automation also extends to data-driven tasks. Multi-agent LLM systems have been applied to process building management system data, predict loads, visualize performance, and generate retrofit recommendations (Xiao and Xu 2024; Xiao and Xu 2024). Applications such as HxAgent autonomously extract mathematical models from engineering literature and produce executable optimization code (Mishra et al. 2025), while frameworks like BrickLLM automate the construction of standardized metadata graphs from natural language descriptions (Perini et al. 2025). Collectively, these approaches reduce reliance on expert-crafted scripts and lower the barriers for professionals without specialized coding expertise.

Nevertheless, automation in domain-specific applications such as BEM represents a paradigm shift. By integrating reasoning with execution, LLM agents minimize

repetitive tasks, improve modelling accuracy, and enable new possibilities for end-to-end intelligent workflows. Insights from cross-domain applications further suggest that the potential of automation is maximized when high-level adaptability is combined with structured protocols and rigorous validation, thereby ensuring that agentic systems can scale safely and effectively into real-world building operations.

These capabilities lay the foundation for more advanced, multi-agent systems capable of executing not only isolated tasks but entire workflows – a topic elaborated in the following subsection.

1.2.2. Introduction to agentic workflows

Driven by advances in Generative AI, the concept of agentic workflows has gradually emerged and attracted increasing scholarly attention. Rather than focusing solely on the functionality of a single agent, agentic workflows emphasize collaboration among multiple agents and tools, enabling the completion of complex tasks through iterative cycles of planning, execution, and verification. Existing research generally categorizes these developments into two levels: (i) the LLM agent, centred on a large language model and capable of autonomously performing limited-scope planning and tool invocation; (ii) the agentic workflow, which highlights multi-step task decomposition and cross-tool collaboration.

These concepts have been applied across a range of domains. For example, in data processing and scientific research, a single LLM agent combined with a code interpreter facilitates rapid computation and visualization. For more complex tasks, multi-agent collaboration divides responsibilities across planning, execution, and verification phases, thereby improving efficiency and reliability. Collectively, these explorations demonstrate that agentic workflows have progressed from theoretical concept to practical application and are becoming an integral component of automation and intelligence in BEM.

Across domains, agent-oriented systems are most suitable for open-ended problems with unpredictable procedural steps, such as automatically generating building energy models or exploring control strategies. Their strength lies in flexible, model-driven decision-making, multi-turn reasoning, memory integration, and adaptive tool use, which make them effective for exploratory tasks and scalable problem solving in trusted environments. However, these advantages come with higher computational cost and increased risk of compounded errors, requiring extensive validation, guardrails, and testing to ensure reliability.

In contrast, workflow-oriented approaches are better suited to well-defined tasks that demand consistency,

predictability, and repeatability. By structuring execution into predefined stages, workflows provide reliable and efficient processing, facilitate validation, and reduce error propagation, thereby ensuring stable and reproducible outputs. Their primary limitation is reduced flexibility, which makes them less appropriate for highly dynamic problems. This distinction is particularly relevant for BEM, where simulation procedures, parameter modifications, and result analyses typically follow deterministic and structured processes. Consequently, workflow-based architectures offer a natural organizational paradigm for coordinating reasoning modules, transformation steps, and simulation tools within such domains. The following sections build on this perspective and examine how these principles can be operationalized within a practical system.

1.2.3. Existing applications of agentic workflows

Although still at an early stage, LLM-based agentic workflows have begun to show practical applications in the BEM field. One notable direction is workflow standardization, where proposals such as JSON-based schemas (Zhang et al. 2025) and open-source agent libraries (Xu et al. 2025) aim to unify how building-related agents are described, shared, and reused, thereby ensuring reproducibility and interoperability across research and practice. These standardization efforts highlight the value of automation not only for individual projects but also for fostering an ecosystem of reusable, domain-specific tools.

Agentic BEM workflows started demonstrating promising potential. For example, Zhang et al. introduced an agentic workflow to automatically generate Energy-Plus models, reducing errors and manual debugging in simulation preparation (Zhang et al. 2025). Zhou et al. combined LLMs with BIM tools such as Revit to streamline 3D model generation from 2D inputs (Zhou et al. 2024), while Yan et al. applied the ReAct strategy for smart building device control, achieving high success rates (Yan et al. 2025). Other contributions include frameworks such as OpenFOAMGPT (Xie and Xu 2023) for automating CFD simulations and BrickLLM (Perini et al. 2025) for semantic metadata construction, both of which reduce labor-intensive steps in modelling and compliance. Together, these studies suggest that BEM is beginning to benefit from agentic designs that improve accuracy, efficiency, and scalability.

Despite these advances, current BEM-oriented agentic workflows face several limitations. Validation practices remain inconsistent, with no standardized benchmarks for assessing reliability across tasks. Scalability is another challenge, as many workflows are computationally intensive and not well-suited for real-time or large-scale deployment. Privacy and security concerns are also

particularly salient, given the sensitivity of building operational data. Finally, the lack of interoperable standards across agents and tools constrains integration and reuse. Addressing these gaps calls for robust contextual protocols, such as the MCP, to enable scalable, secure, and interoperable deployment of LLM-driven agentic workflows in practice.

1.3. Model context protocol for enabling AI interactions with BEM

This section reviewed how recent GenAI advancements (e.g. reasoning, tool use, code execution) make it feasible to automate the traditionally labor-intensive BEM tasks. Existing agentic BEM workflows still require developers to create bespoke connectors for each data source, tool, and service, leading to a tangle of one-off integrations. A key reason behind this is the lack of a standardized way for LLMs to reliably access those external resources. To resolve this kind of challenge, Anthropic introduced the Model Context Protocol (MCP) in late 2024. Essentially, MCP is the ‘USB-C for AI apps’ that provides a universal interface that allows LLMs to read and write files, call APIs/functions, query databases, and generally interact with the outside world in a standardized way (What is the Model Context Protocol (MCP)? 2025).

Since launch, the ecosystem has grown rapidly, with official SDKs (Model Context Protocol 2025), example servers, and a growing registry that simplifies discovery and reuse. Notable adoptions span diverse domains:

- **Software Development:** Major IDE platforms and command line coding assistants like VSCode, Cursor, Codex, and Claude Code have integrated MCP to provide AI coding assistants with real-time access to project context, enabling features like automated debugging and code generation.
- **Enterprise Productivity:** Organizations use MCP servers for Google Drive, Slack, and Gmail to enable AI assistants to search documents, manage communications, and automate workflows. Block and Apollo have integrated MCP into their systems for business process automation (Block, Inc 2025).
- **Research and Knowledge Management:** Academic tools like Zotero employ MCP for semantic literature searches, PDF annotation extraction, and AI-assisted review generation, demonstrating the protocol’s value in knowledge-intensive workflows (Yu 2025).
- **Data and Analytics:** MCP servers for PostgreSQL, vector databases, and analytics platforms enable natural language querying and automated data pipeline construction, bridging the gap between technical infrastructure and business users.

In the building domain, MCP adoption is emerging as organizations explore how AI agents can interrogate Building Information Models (BIM), automate coordination tasks, and bridge design-to-simulation workflows. Autodesk demonstrated a ‘conversational BIM’ prototype where an MCP server connects AEC data APIs with a 3D viewer, enabling natural language queries and dynamic model highlighting (Talk to Your BIM: Exploring the AEC Data Model with MCP Server + Claude and Autodesk Platform Services 2025). This implementation allows stakeholders to interact with complex building models through conversational interfaces rather than traditional technical commands. Academic research on multi-agent design, such as the Text2BIM framework (Du et al. 2025), illustrates emerging workflow patterns where specialized agents parse natural language instructions, plan spatial layouts, invoke BIM APIs, and iterate with rule-based validation checks (Talk to Your BIM: Exploring the AEC Data Model with MCP Server + Claude and Autodesk Platform Services 2025). While not yet standardized on MCP, such work demonstrates the architectural patterns that MCP could formalize across the industry. Industry analysis notes that while AEC’s tight project schedules and entrenched legacy tools slow experimentation, MCP significantly lowers integration costs and helps transition from proof-of-concept demonstrations to sustained operational practice (VIKTOR.AI 2025).

Within BEM specifically, formal MCP applications remain extremely limited. The EnergyPlus-MCP is a pioneering effort to expose a comprehensive simulation engine’s capabilities through a standards-based interface (Li, Xu, and Hong 2025). This MCP server provides programmatic access to the full spectrum of BEM workflows: model loading and validation, parametric configuration, simulation execution control, and results inspection and analysis. Through 35 specialized tools spanning model management, HVAC system configuration, and post-processing capabilities, the server establishes foundational infrastructure that enables any MCP-compatible client to interact with EnergyPlus through natural language interfaces. While this technical foundation exists, however, the field lacks concrete demonstrations of how these capabilities translate into practical agentic workflows for common BEM tasks.

This gap between available MCP infrastructure and demonstrated BEM applications motivates our current work. While the EnergyPlus-MCP server provides the necessary tools and interfaces, practitioners need systematic exploration of how these capabilities enable agentic workflows for model calibration, retrofit analysis, parametric studies, and optimization tasks.

1.4. Objectives and contributions of this research

This research proposes a novel framework of MCP-enabled agentic AI workflows to systematically explore the applications of generative AI in BEM. The objectives include developing an agentic AI system that leverages natural language interaction with simulation tools to automate model inspection, as well as achieving end-to-end automation of simulation tasks encompassing execution, results collection, and quality controls. The framework is assessed for its potential in scalability, cross-platform interoperability, and BEM education or training, thereby advancing BEM toward greater ubiquity/adoption and effective use.

Major contributions of this research include:

- Introduction of a novel BEM framework with two complementary paradigms of agentic workflows to automate BEM tasks with AI agents, from single-agent conversational interactions to multi-agent autonomous workflows, each requiring different orchestration strategies, memory handling, and levels of human involvement.
- Illustration and evaluation of the two agentic BEM workflow: conversational integration for exploratory tasks, and agentic workflow integration for systematic automation.
- Demonstration of how AI agents can leverage MCP to navigate complex modelling decisions, execute iterative refinements, and synthesize results across multiple scenarios.

The rest of this paper is organized as follows. Section 2 describes the methodology on MCP and agentic AI and how they map to the BEM workflow, as well as evaluation metrics. Section 3 presents the use cases and illustrates two examples on model inspection, edit, and simulation, and sensitivity analysis with MCP-integrated Agentic Workflow. Section 4 further discusses major contributions, limitations and future research opportunities. Finally, Section 5 concludes the study.

2. Methodology

2.1. Mcp integration paradigms

This study explores two complementary paradigms for integrating EnergyPlus-MCP into practical workflows, as illustrated in Figure 1. First, a conversational integration places a single agent inside a chat-centric client (e.g. Claude, VS Code) where the user iteratively steers the session and the agent invokes EnergyPlus-MCP and optionally other MCP tools on-demand. Second, an

agentic workflow integration coordinates a set of specialized agents (e.g. planner, model inspector, simulation manager, calibration expert) that autonomously call EnergyPlus-MCP and optionally other tools while maintaining state and memory to produce end-to-end outputs. Both paradigms leverage MCP's standardized tool interface but differ in their interaction styles, memory handling, and level of autonomy.

2.1.1. Conversational integration

The conversational integration paradigm places EnergyPlus-MCP inside a chat-centric client where a single agent collaborates with the user through natural language. Figure 2 depicts four swim lanes for the User, the MCP client, the EnergyPlus-MCP server, and optional MCP servers that provide domain knowledge or benchmarks. A session typically begins when the user prepares an EnergyPlus model and a weather file, then poses high-level questions about the model. The MCP client interprets intent, selects appropriate tools, packages inputs as JSON, and calls the server. The server executes inspection functions and returns structured JSON. The client parses these results and summarizes the key facts in plain language so the user can decide what to do next.

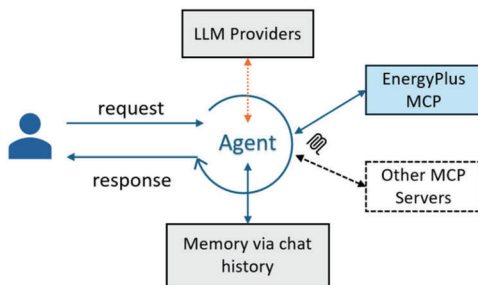
When model edits are requested, the client infers intent, chooses the editing tools shown in Figure 2, prepares JSON inputs, and calls the server. The server applies the requested changes and returns JSON describing what was modified. The client turns this into an auditable narrative, which lets the user confirm the changes or request further adjustments. If the user needs design guidance, the client can consult an external knowledge server such as a BEM encyclopedia and incorporate the retrieved guidance into the next tool call.

The workflow is not a fixed sequence. Dashed connectors in Figure 2 indicate optional branches and loops. The user can move directly from model inspection to simulation without intermediate edits. After a simulation, the user can return to inspection or editing based on observed results. This flexible interaction is deliberate and supports rapid exploration. The conversation history preserves context and intent across turns, which enables continuity of decisions and provenance of changes while the agent handles tool selection, JSON input and output, and orchestration of EnergyPlus-MCP functions.

2.1.2. Agentic workflow integration

The agentic workflow integration paradigm organizes EnergyPlus-MCP into a coordinated system of specialized agents that work with minimal human intervention. Instead of a single conversational agent interpreting user queries, this approach employs distinct roles such as a planner, model inspector, simulation manager, calibration expert, error analyzer, and result analyzer. Each agent is responsible for a specific function and collaborates with others to complete tasks in sequence or in parallel. The planner decomposes a high-level objective into actionable steps, which are then delegated to the appropriate agents. These agents call EnergyPlus-MCP and other MCP-enabled servers as needed, maintain explicit state and memory, and exchange intermediate results until a final outcome is produced. Human input is typically limited to setting the overall objective and reviewing outputs such as calibration reports, benchmarking analyses, or compliance checks. This paradigm emphasizes autonomy, efficiency, and scalability, making it well suited for complex or repetitive BEM tasks where human guidance can be minimized in favour of automated execution.

(a) Conversational Integration



(b) Agentic Workflow Integration

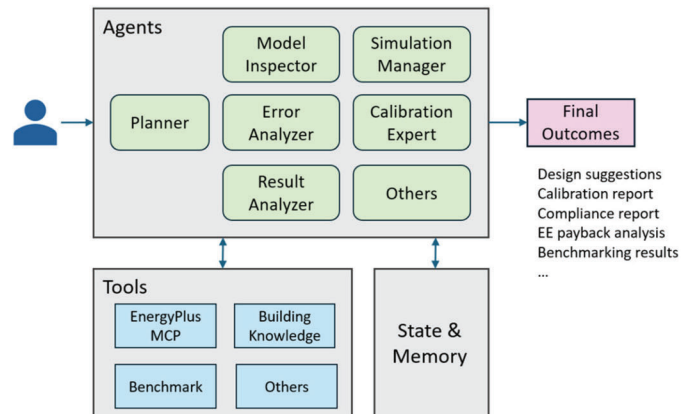


Figure 1. Two paradigms for integrating EnergyPlus-MCP: (a) conversational integration in a chat-based environment and (b) agentic workflow integration with multiple specialized agents.

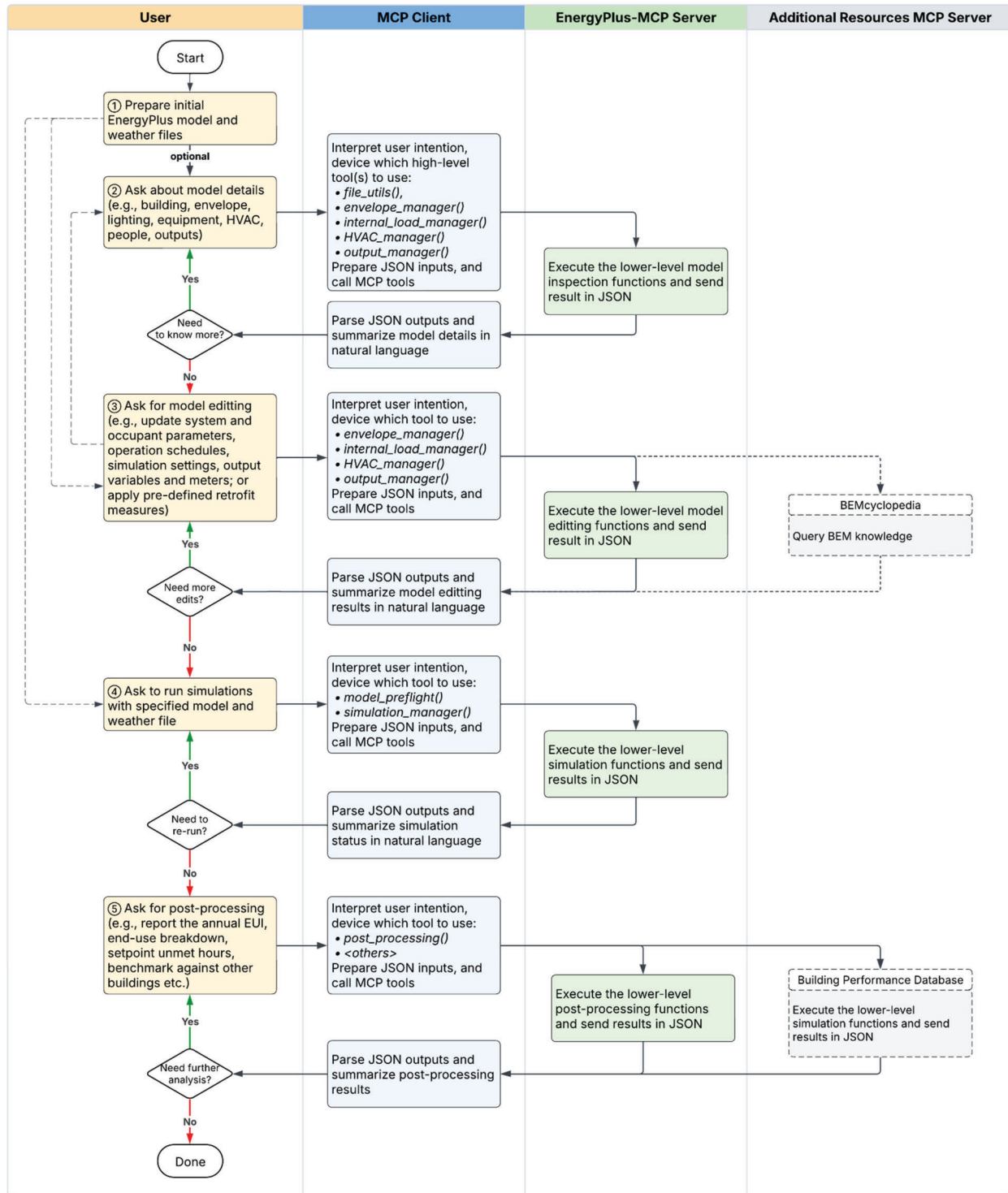


Figure 2. Swimlane workflow of conversational integration, showing how the user, MCP client, EnergyPlus-MCP server, and optional MCP servers interact in flexible, iterative loops for model inspection, editing, simulation, and post-processing.

2.2. Representative BEM tasks

To evaluate and compare the aforementioned conversational and agentic integration paradigms, we design a set of representative BEM tasks that span the major activities in a typical workflow. These tasks are grouped into five categories: inspection and query, editing and simulation,

debugging and repair, retrofit analysis, and model calibration. Within each category, tasks are structured at three difficulty levels so that we can observe how task complexity influences user effort, automation potential, and task completion quality. The difficulty progression is based on factors such as the number of steps involved, the degree of reasoning required, and the need for model

Table 1. Representative BEM tasks used for comparison of conversational and agentic integration paradigms.

Category	Difficulty	Tasks
Inspect and Query	Easy	List all zones or schedules referenced by a specific thermostat.
	Medium	Trace dependencies for a named coil or schedule across multiple objects.
Edit and Single Simulation	Easy	Change a cooling setpoint schedule or swap a material property.
	Medium	Add an infiltration schedule to a zone and update related lists.
	Hard	Insert a new thermal zone with surfaces, constructions, and loads consistent with geometry and adjacency.
Debug and Repair	Easy	Fix a missing schedule reference that prevents simulation.
	Medium	Resolve conflicting node names across an HVAC loop so the model runs cleanly.
	Hard	Reconfigure auto-sized components with incompatible control schemes into a coherent plant loop.
Retrofit Analysis	Easy	Adjust a thermostat setpoint to evaluate potential savings.
	Medium	Apply an envelope or lighting retrofit measure and compare energy use with the baseline.
	Hard	Conduct multi-measure retrofit analysis and generate comparative performance reports.
Model Calibration	Easy	Calibrate a model with one parameter tweaked, with the parameter name provided (e.g. internal load multiplier).
	Medium	Calibrate a model with two or three parameters tweaked, with parameter names provided (e.g. infiltration rate, lighting power density, occupancy schedule).
	Hard	Calibrate a model with multiple parameters tweaked, without revealing which ones, and match baseline performance within MAPE and CVMSE thresholds.

modifications or external knowledge. Table 1 summarizes the selected tasks that will be used for evaluation.

This structure ensures that the tasks are clear, reproducible, and representative of real-world BEM workflows. This task set is not intended to be exhaustive. Instead, it provides a representative sample that establishes a testing and evaluation framework for comparing integration paradigms. The framework can be expanded in future work to include additional categories and tasks as new BEM use cases emerge.

2.3. Scope and evaluation considerations

Comprehensive evaluation of MCP integration paradigms would involve metrics including efficiency (time to completion, token consumption), effectiveness (task correctness, tool precision, error recovery), and human effort (user interventions, interaction turns). Such benchmarking requires standardized test models, controlled experimental conditions, and multiple trials to account for LLM response variability. Additionally, workflow quality is influenced by both the LLM's reasoning and tool-calling capabilities and the user's ability to formulate effective queries; these dependencies differ across integration paradigms, as discussed in Section 4.2.

This paper does not attempt such quantitative benchmarking. Instead, our focus is on demonstrating the feasibility and workflow characteristics of two MCP integration paradigms through end-to-end use cases. Section 3 presents these demonstrations with practitioner-informed estimates of time savings to illustrate practical benefits, while acknowledging that rigorous measurement under controlled conditions warrants a dedicated study. This scope is appropriate for introducing a new integration approach, where establishing workflow patterns and surfacing practical considerations provides the necessary foundation for subsequent quantitative evaluation.

3. Use cases and demonstrations

3.1. Building and model description

The demonstration employs the iUnit, an experimental micro-apartment prototype developed by the National Renewable Energy Laboratory (NREL, now NLR) in Golden, Colorado. Designed as a modular and transportable residential system, the iUnit provides a well-documented and physically transparent platform for building energy simulation (Figure 3). The model represents a compact single-occupancy dwelling comprising three thermal zones – a main living area, a bathroom, and a service/storage zone – capturing the essential spatial and thermal interactions typical of small urban apartments while maintaining a manageable simulation scale.

The envelope system features lightweight wall and roof constructions with defined insulation layers, coupled with a single south-facing window that facilitates controlled daylight and solar heat gain analysis. This configuration allows clear attribution of thermal responses to envelope and system parameter variations, making the iUnit particularly suitable for agent-driven inspection, modification, and optimization workflows.

All simulations were executed under consistent boundary conditions using the TMY3 weather file for Golden, Colorado to ensure comparability and reproducibility across model interrogation and sensitivity analysis tasks. The iUnit thus serves not merely as a geometric test case but as an experimental reference platform – ideal for evaluating the transparency, adaptability, and reasoning capability of the MCP-integrated AI assistant in model interpretation, editing, and automated sensitivity analysis.



Figure 3. The NREL iUnit experimental testbed used for case study: (a) exterior view of the modular facility in Golden, Colorado; (b) schematic floor plan showing three thermal zones and south-facing window configuration.

The selection of a compact model is methodologically deliberate: this study demonstrates that MCP's standardized interface supports architecturally distinct integration paradigms rather than stress-testing tool performance against model complexity. A transparent building with well-characterized thermal behaviour enables clear attribution of workflow patterns without confounding effects, and using the same model across both paradigms enables controlled comparison under identical conditions. The present study demonstrates end-to-end workflows that orchestrate multiple tools within two distinct architectural paradigms, evaluating workflow-level characteristics including time efficiency, reproducibility, and transparency.

3.2. Model inspection, edit, and simulation with MCP-connected AI assistant

As described in Section 2.1.1, conversational integration enables users to interact with EnergyPlus through natural language by connecting an AI client to the EnergyPlus-MCP server. In this demonstration, we use Claude Desktop as the MCP client, though other clients such as GooseAI, VSCode, and similar platforms can serve the same role. The client interprets user intent, invokes appropriate MCP tools, and synthesizes results into comprehensible responses. To demonstrate this paradigm, we present a scenario where a building energy consultant receives the iUnit baseline model with three objectives: (1) inspect the model to understand its characteristics, (2) make modifications to evaluate design alternatives, and (3) run simulations to compare energy performance. The following subsections demonstrate this workflow through comprehensive model inspection, modification using two complementary approaches, automated simulation and performance comparison, and a summary of key advantages over traditional manual workflows.

3.2.1. Model inspection

The consultant/user began by exploring available MCP capabilities and confirming that the iUnit model and weather files were accessible in the workspace. The *file_utils* tool verified file locations, establishing readiness for inspection tasks.

Comprehensive Summary Through Tool Orchestration. When asked to provide a building summary, the AI assistant orchestrated multiple MCP tools (*model_pre-flight*, *hvac_manager*, and *internal_load_manager*) in response to a single natural language query. This coordination demonstrates a key advantage of conversational integration: rather than invoking each tool separately, users express intent naturally while the AI assistant manages MCP tool selection and execution. The resulting summary synthesized information across multiple domains: building location and climate (Golden, Colorado at 1,829 m elevation), geometry (29 m² floor area divided into three zones), envelope construction (high-performance walls and roof with R-23 and R-40 insulation respectively), HVAC configuration (no centralized systems, suggesting passive or naturally ventilated operation), and internal loads (lighting and occupant specifications for each zone). This comprehensive overview, obtained in around 30 s, replaced the 20-30 min typically required to manually parse IDF files and cross-reference object definitions.

Interactive Building Geometry Visualization. The consultant requested a 3D visualization to understand building geometry and zone layout. The *geometry_manager* tool generated an interactive HTML viewer embedded directly in the conversation. Figure 4 shows the generated 3D visualization with building information panel, interactive controls for rotation and panning, and customizable visualization settings including visibility toggles and transparency sliders for different surface types. The L-shaped living room configuration is clearly visible along with the adjacent bathroom and closet zones. This

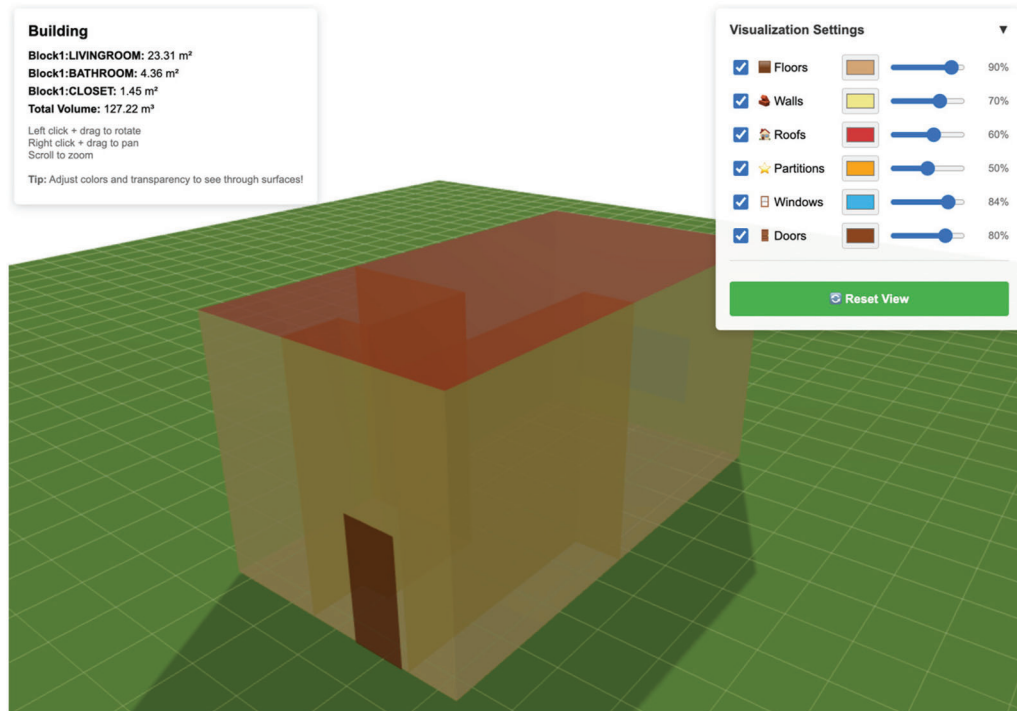


Figure 4. 3D Geometry visualization HTML dynamically generated by the MCP tool.

artifact generation capability reduced what would traditionally require 15–20 min with SketchUp or 30–45 min with custom Python visualization scripts to under 2 min.

Envelope Inspection and Cleanup. The consultant then requested detailed envelope analysis to understand construction assemblies and material properties. The *envelope_manager* tool generated an interactive HTML interface presenting the complete envelope hierarchy: zones, surfaces, constructions, materials, and their relationships. Figure 5 illustrates this interface, with the left panel showing construction and material properties organized by zone, and the right panel highlighting envelope quality issues. The visualization immediately revealed 6 orphaned constructions (constructions defined but not assigned to any surface) and 1 orphaned material, representing objects that added complexity without serving any functional purpose in the model. Based on this discovery, the consultant requested cleanup of these unused components. The AI assistant used the MCP tool to remove the orphaned objects and 1 additional material that was only used within an orphaned construction, creating a cleaned model with 7 constructions instead of the original 13, and 16 materials instead of 18. A subsequent verification confirmed successful cleanup, with the cleaned model retaining only components actively used by building surfaces. This relationship tracking capability, which identifies unused objects and their cascading dependencies, would be

extremely tedious to perform manually through text-based IDF editing.

Internal Loads and HVAC Systems. A subsequent query about lighting, equipment, and HVAC provided detailed specifications for lighting power densities (5–7.5 W/m² across zones, totalling 156 W), occupancy patterns with zone-specific schedules, and confirmed the absence of electric equipment and centralized HVAC systems.

The complete inspection workflow, conducted through approximately five natural language exchanges in less than 10 min, provided comprehensive model understanding across geometry, envelope, loads, and systems. This represented a significant time reduction compared to traditional manual inspection (which typically consumes 1 to 2 h). The conversational approach enabled progressive exploration where each insight informed subsequent queries, while maintaining full transparency through visible tool invocations and auditable results.

3.2.2. Model modification for energy retrofit

The modification phase focused on envelope improvement, demonstrating how conversational integration enables systematic retrofit planning through model inspection, measure selection, and alternative scenario creation.

Building Envelope Assessment and Retrofit Opportunities. The consultant first requested comprehensive

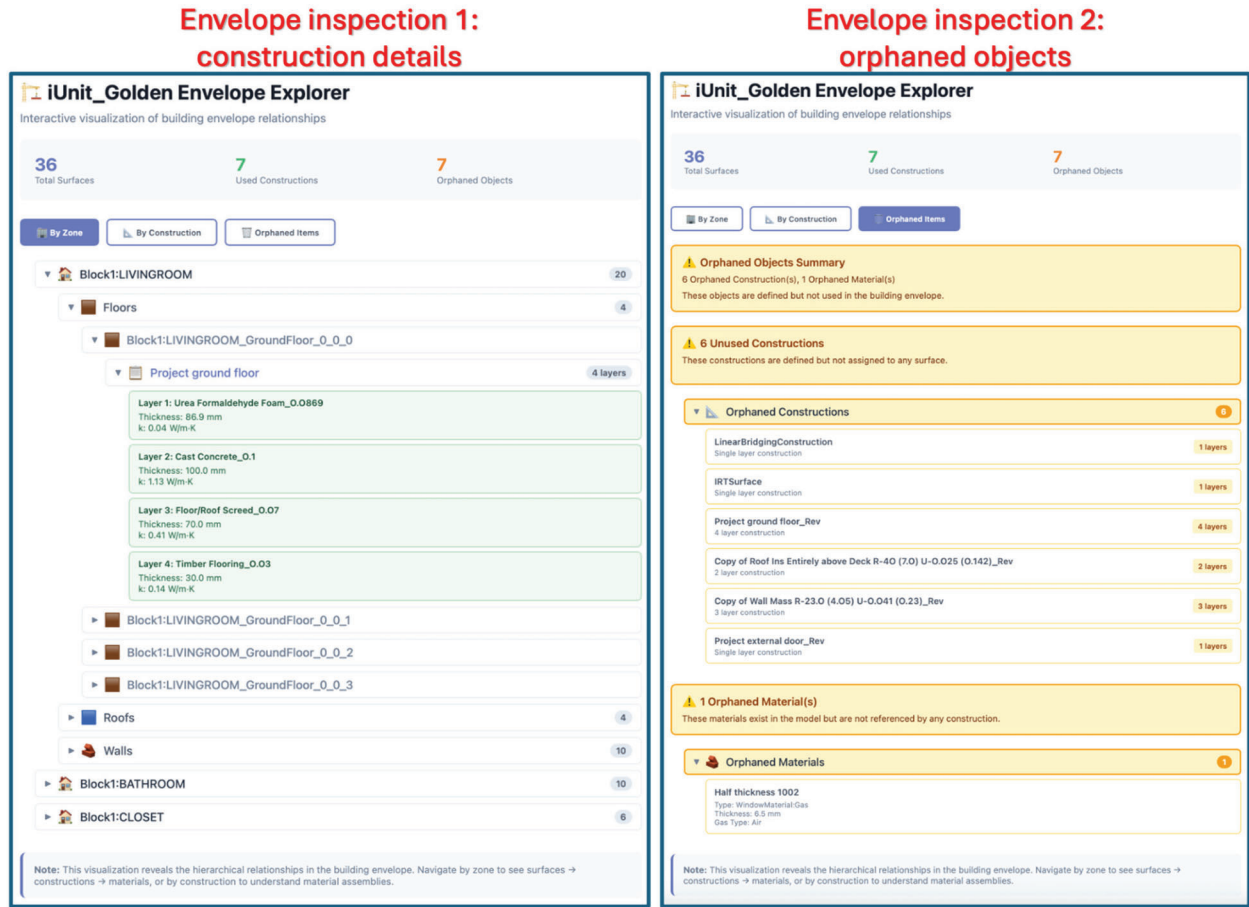


Figure 5. Envelope inspection HTML dynamically generated by the MCP tool. Left: construction and material properties details organized by zone; Right: orphaned objects that are not used by any surface.

Table 2. Envelope performance analysis and improvement priorities.

Component	Current R-Value	Code Requirement (CZ5)	Status	Priority
Exterior Walls	R-23.8	R-20 to R-30	Meets/Exceeds	Low
Roof	R-39.2	R-49 minimum	Below Code	High
Floor	R-15.0	R-30 minimum	Below Code	High
Windows	U-0.28, SHGC-0.50	$U \leq 0.30$, any SHGC	Meets Code	Low
Entry Door	R-1.0	R-5 to R-10 typical	Poor	Medium

envelope inspection to identify improvement opportunities. The AI client analyzed envelope components against Climate Zone 5 code requirements using web search augmentation, evaluating current performance and prioritizing potential retrofits. Table 2 summarizes this analysis.

The analysis revealed that while walls met requirements, the roof assembly fell short of code minimum and floor insulation provided only half the required R-30 value. Windows met code despite using air gaps rather than more efficient argon gas fill, presenting a cost-effective improvement opportunity. This assessment, completed in 5-7 min through conversational queries, would traditionally require 45-60 min of manual calculations and code reference lookups.

Retrofit Implementation Approaches. The consultant explored two complementary modification app

roaches. First, using the retrofit_manager tool to discover pre-defined energy conservation measures, which revealed four ready-to-use options: high-performance window film, cool roof coating, cool wall coating, and infiltration reduction. The AI client analyzed each measure's climate appropriateness for Golden, Colorado's heating-dominated climate (ASHRAE Climate Zone 5B). Infiltration reduction received high priority given cold winter conditions where air leakage represents the primary heat loss pathway, with potential savings of 10%–30% in heating energy. Conversely, the three reflective coating measures were flagged as lower priority or counterproductive, as reducing solar heat absorption would increase winter heating loads. This climate-aware analysis enabled informed decision making where AI recommendations augmented professional judgment.

Second, the consultant employed custom modification tools (*envelope_manager* and *idf_modification*) to address envelope gaps not covered by pre-defined measures. For window improvement, the *idf_modification* tool replaced air gaps with argon gas in the triple-pane construction, reducing thermal conductivity and improving U-factor by 15-20%. For roof upgrade, the *envelope_manager* tool increased insulation thickness to achieve R-60, exceeding code minimum to account for thermal bridging. These custom modifications demonstrate the precision available when pre-defined measures do not address specific retrofit requirements.

Alternative Scenario Creation. The consultant created three progressive retrofit scenarios. Table 3 summarizes the baseline model and three alternatives.

Alternative 1 applied the infiltration reduction measure using *retrofit_manager* with a 50% multiplier, systematically modifying all zone infiltration objects. Alternative 2 extended this with window improvement through *idf_modification*, replacing air with argon gas to enhance thermal performance while maintaining beneficial solar heat gain. Alternative 3 provided a comprehensive envelope package by adding roof insulation upgrade through *envelope_manager*, achieving code compliance and providing performance margin. Each alternative was saved with descriptive naming conventions (*alt1_infiltration*, *alt2_infiltration_windows*, *alt3_full_package*) for clear identification during simulation analysis. This staged approach enables comparison of incremental versus comprehensive retrofits to support cost-benefit evaluation.

The complete workflow, from envelope assessment through creation of three alternative scenarios, required 15-20 min through conversational interaction. Traditional approaches would require 45-60 min for envelope analysis plus 30-45 min per alternative for manual IDF editing (90-135 min for three models), totalling 135-195 min. The conversational approach achieved 85%–90% time reduction while maintaining full transparency through visible tool invocations and systematic documentation of modification decisions.

3.2.3. Simulation and performance comparison

Having created baseline and alternative models, the consultant proceeded to simulations and compare retrofit outcomes. This phase demonstrates automated simulation

execution, results extraction, and AI-generated visualization with embedded recommendations.

Automated Simulation Execution. The consultant requested parallel simulation of all four models using the Golden, Colorado weather file. The *simulation_manager* tool executed simulation runs sequentially, completing in approximately 16 s total (3-5 s per model). Each simulation generated comprehensive outputs including SQLite databases, CSV time-series files, HTML reports, and error logs, automatically organized into separate directories for each scenario. This automated execution contrasts with traditional approaches requiring manual simulation setup, output directory configuration, and individual run monitoring.

Results Extraction and Annual Energy Comparison. The consultant queried available outputs using the *post_processing* tool, which revealed 101 unique variables tracked at multiple reporting frequencies across 8,760 annual timesteps. The tool extracted energy consumption data for electricity, district heating, and district cooling across all scenarios. The comparison showed that district heating dominated total energy use, making envelope improvements highly impactful in this heating-dominated climate. Alternative 1 (infiltration reduction) achieved 20.9% total energy savings, demonstrating that air sealing provides the majority of achievable benefits. Alternative 2 (adding window argon upgrade) provided modest incremental savings of 1.0%, while Alternative 3 (comprehensive package) achieved 24.6% total savings. The analysis also revealed a cooling penalty where tighter envelopes increased cooling loads by 11-16%, though heating savings far outweighed this effect. This extraction and comparison, accomplished through conversational queries, would traditionally require custom SQL queries or Python scripts to parse files and calculate statistics.

Interactive Visualization and Recommendations. To better understand seasonal patterns and cost-effectiveness, the consultant requested monthly energy comparison across all scenarios. The AI client generated an interactive HTML artifact combining data visualization with analytical insights. Figure 6 presents this artifact, featuring monthly heating and cooling consumption line charts for all four scenarios, annual summary cards showing percentage changes, and an integrated panel with

Table 3. Retrofit scenario comparison.

Scenario	Description	Infiltration	Window U-factor	Roof R-value	MCP Tools Used
Baseline	Current model	Standard (ACH = 1.0)	U ~ 0.28	R-39.2	–
Alt 1	Air sealing only	50% reduction	U ~ 0.28	R-39.2	<i>retrofit_manager</i>
Alt 2	Air sealing + windows	50% reduction	U ~ 0.24 (argon)	R-39.2	<i>retrofit_manager idf_modification</i>
Alt 3	Comprehensive package	50% reduction	U ~ 0.24 (argon)	R-60	<i>retrofit_manager idf_modification envelope_manager</i>

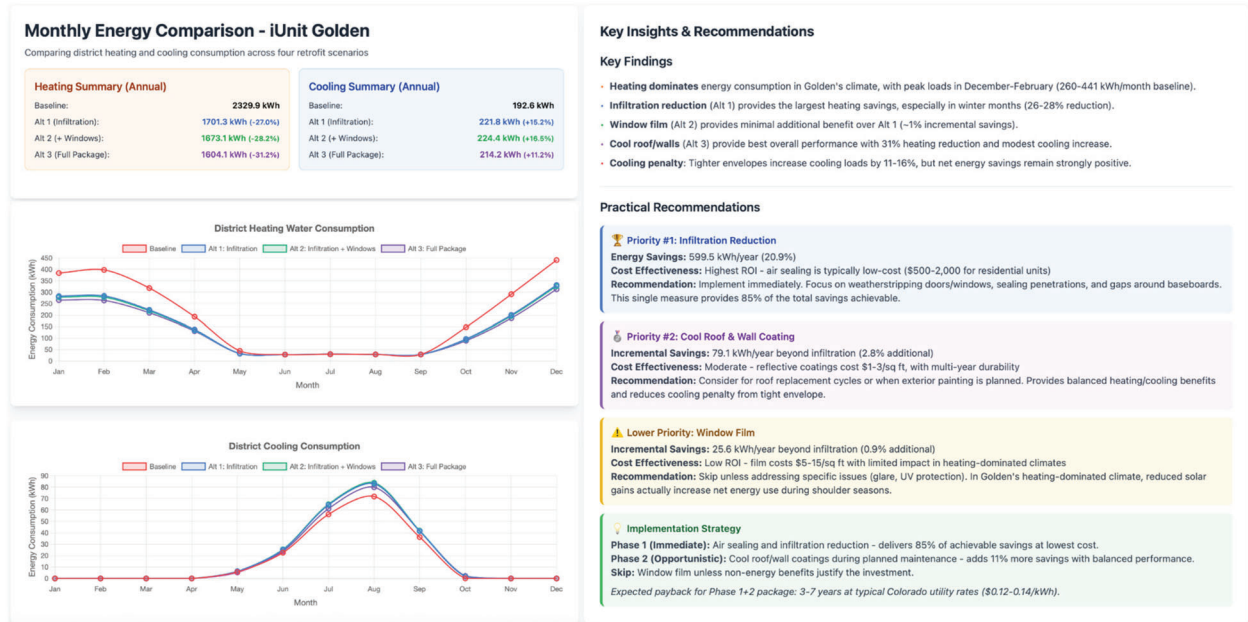


Figure 6. Interactive HTML showing energy performance comparison and retrofit recommendations.

key findings and practical recommendations. The visualization clearly shows winter heating peaks (December-February at 260–441 kWh/month baseline) where infiltration reduction provides maximum benefit, and modest summer cooling loads where envelope improvements show smaller incremental effects. Beyond data visualization, the AI client synthesized cost-effectiveness recommendations directly from simulation results, prioritizing infiltration reduction as the highest ROI measure (delivering 85% of achievable savings at lowest cost), recommending cool roof and wall coatings as opportunistic improvements during planned maintenance, and flagging window film as lower priority given minimal incremental benefit in heating-dominated climates. This integration of simulation data, visualization, and actionable recommendations in a single artifact exemplifies how AI-assisted workflows extend beyond data processing to decision support. The artifact generation capability reduced what would traditionally require separate Python plotting scripts, manual calculation of percentage changes, and written interpretation of results into a single conversational request producing publication-ready output.

The complete simulation and analysis workflow, from parallel execution through interactive visualization with embedded recommendations, required 10-15 min through conversational interaction. Traditional approaches would require hours of manual efforts: 15-20 min for manual simulation setup and monitoring, 30-60 min for results extraction through SQL queries or CSV parsing scripts, 30-60 min for creating plots with Python or R

visualization libraries, and 10-20 min for calculating summary statistics and writing interpretive text. The conversational approach achieved significant time reduction while producing high-quality outputs through AI-synthesized insights and integrated artifact generation that combines data, visualization, and recommendations in coherent narratives rather than disconnected technical outputs.

3.3. Automatic model-based optimization with MCP-integrated agentic workflow

To demonstrate the operational efficacy of the MCP-based BEM framework, this section details an automated model-based optimization case study. We implemented an end-to-end application leveraging crewAI flow for multi-agent orchestration, integrating local reasoning agents with remote MCP-enabled simulation tools. This proof-of-concept illustrates how the framework bridges the gap between high-level architectural intent and low-level engineering rigour, enabling non-specialists to perform systematic optimization while ensuring computational transparency.

3.3.1. Workflow overview

Building upon the iUnit benchmark introduced in Section 3.1, we developed a crewAI-based automated agentic workflow that integrates LLM reasoning, rule-based parameter transformation, and containerized EnergyPlus simulations through the MCP. The workflow shown in Figure 7 operates as a multi-stage computational pipeline, where each stage functions as an autonomous agent executing well-defined transformations of the

building model state. The diagram highlights the alignment between the local agent structure and the corresponding MCP tools – inspect, preflight, simulation, and visualize, which together enable a fully traceable optimization cycle from model interrogation to performance interpretation.

The workflow adopts a hybrid architecture that combines LLM-driven reasoning for tasks requiring judgment (e.g. parameter identification, result interpretation) with rule-based transformation for deterministic operations (e.g. geometry scaling, material property modification). Different LLMs are assigned to distinct reasoning roles

within the workflow. GPT-5 is employed during the parameter schema construction stage (Section 3.3.3) to perform high-level structural reasoning and constraint interpretation due to its stronger generalization and abstraction capabilities. In contrast, GPT-4o-mini is used during the result interpretation and sensitivity analysis stages (Section 3.3.5), where repeated analytical reasoning is required, to reduce computational cost without affecting correctness. This tiered model allocation balances reasoning capability and computational efficiency, preventing unnecessary use of high-capacity models in repetitive tasks.

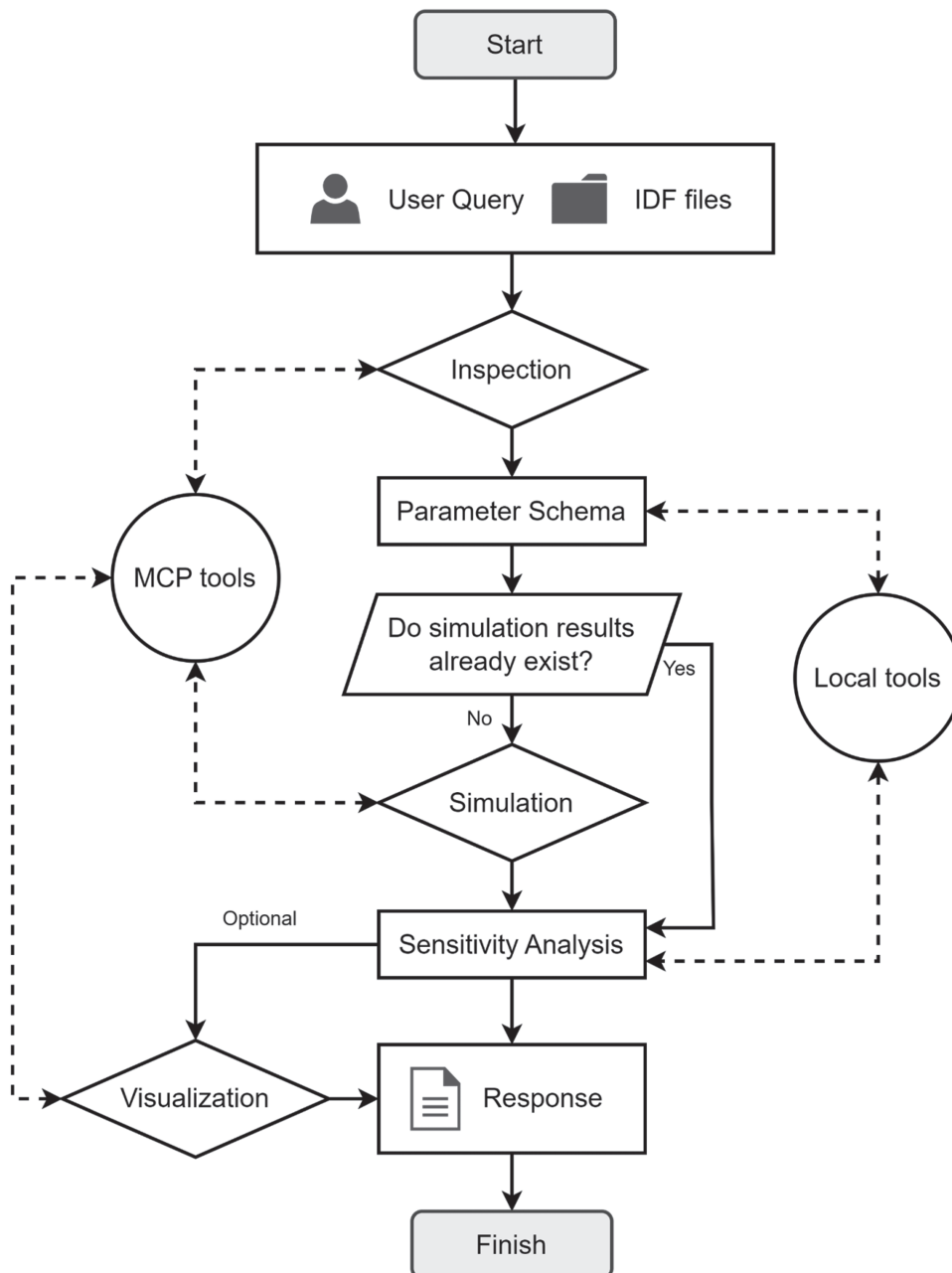


Figure 7. MCP-integrated agentic workflow for automatic building energy model.

The workflow begins with a requirement analysis phase, where the system parses user-specified optimization objectives and constraints. Subsequently, the workflow leverages the MCP inspect tool to scan the user-uploaded baseline IDF file, extracting fundamental building characteristics including geometry, construction assemblies, thermal zones, and HVAC system configurations. This initial inspection provides the foundational context for subsequent optimization decisions.

In the parameter identification and optimization phase, the integrated LLM analyzes the baseline model to identify modifiable building envelope parameters, such as wall insulation thickness, window U-values, glazing properties, and roof construction materials. The LLM then proposes feasible optimization ranges for each identified parameter based on building codes, physical constraints, and energy performance considerations. This stage generates structured feedback and produces multiple modified IDF variants, each representing a specific parametric configuration.

The simulation execution phase employs the MCP simulation tool to sequentially run EnergyPlus simulations for both the baseline IDF and each parametrically modified variant. This batch simulation process yields comprehensive energy performance metrics, including heating/cooling loads, energy consumption profiles, and thermal comfort indicators for each configuration.

Finally, the workflow conducts automated sensitivity analysis by systematically comparing the simulation results across all parametric variations. The MCP plot tool generates visualization outputs, like comparative energy consumption charts, which enable users to readily interpret the optimization outcomes and identify the most effective envelope modifications.

3.3.2. MCP-based model inspection

The workflow begins with a comprehensive model-inspection phase, where the baseline iUnit EnergyPlus model (introduced in Section 3.1) undergoes systematic validation and metadata extraction within an MCP-controlled containerized environment. As summarized in Table 4, five MCP tools are orchestrated to validate model integrity, extract envelope and system information, and ensure simulation readiness prior to automated modification. This inspection leverages five specialized

MCP tools including `model_preflight`, `envelope_manager`, `internal_load_manager`, `hvac_manager`, and `outputs_manager` to establish simulation readiness and extract hierarchical building information, and detailed inspection outputs are provided in **Appendix A1**.

The `model_preflight` tool confirms the structural validity of the baseline IDF file (`/workspace/user-models/iUnit_Golden.idf`) against the EnergyPlus v25.1 IDD schema, verifying weather file references and simulation control settings with no critical errors detected. This step ensures consistent downstream execution within the MCP simulation environment.

Envelope inspection performed by the `envelope_manager` identifies 34 total surfaces (22 exterior and 12 interior), associated with 13 construction assemblies and 18 material definitions. Eight constructions are found to be orphaned, revealing opportunities for envelope simplification and targeted parameter modification. Complementary inspection using the `internal_load_manager` and `hvac_manager` indicates that the model relies on ideal loads assumptions, with three thermal zones defined and no explicit air or plant loop systems. This modelling choice motivates the workflow's focus on envelope-level parameters rather than mechanical system optimization.

Finally, the `outputs_manager` enumerates available metres and output variables, confirming that annual heating and cooling loads, peak demand metrics, and zone-level performance indicators can be consistently extracted for all simulation variants. Together, these inspection results establish both the baseline modelling assumptions and the feasible parameter space for subsequent envelope-level optimization, directly informing the parameter schema construction described in the following section.

In conventional manual workflows, extracting comparable building geometry, envelope composition, and system-level information from an EnergyPlus IDF typically requires domain expertise and approximately 10 min of manual inspection and documentation. In contrast, when executed through the MCP-orchestrated inspection stage, the same information is extracted automatically within approximately 1–2 min, substantially reducing expert effort while ensuring consistent coverage.

Table 4. MCP-based model inspection tools and case-specific outputs.

MCP Inspect Tool	Scope	Key Outputs	Role in Workflow
<code>model_preflight</code>	IDF integrity & schema	IDD version, errors, site info	Simulation readiness
<code>envelope_manager</code>	Envelope geometry & materials	Surfaces, constructions, materials	Identify envelope parameters
<code>internal_load_manager</code>	Occupancy & loads	Schedules, plug loads	Context for thermal demand
<code>hvac_manager</code>	HVAC configuration	Ideal loads, system absence	Define modelling assumptions
<code>outputs_manager</code>	Output variables	Metres, CSV/ESO availability	Enable downstream analysis

3.3.3. Parameter schema construction

Following baseline characterization, the workflow initiates a parameter schema construction phase, in which an LLM-based reasoning agent translates user-specified optimization objectives into a constrained set of actionable envelope-level modifications. Given a goal such as ‘reduce cooling load while maintaining comfort’, the agent interprets inspection results and formulates parameter adjustments grounded in fundamental building physics.

To ensure architectural relevance and physical feasibility, the parameter search space is explicitly constrained through a prompt-embedded whitelist that limits candidate variables to envelope-related components, including exterior walls, windows, roofing assemblies, and building orientation. This design choice emerged from iterative testing, where unconstrained LLM reasoning occasionally proposed physically or programmatically invalid actions, such as altering thermal zone floor areas to reduce cooling demand. By restricting modifications to parameters typically adjustable during early-stage architectural design, the workflow balances the flexibility of LLM reasoning with the need for context-aware and practically implementable optimization strategies.

Under a traditional manual workflow, identifying feasible envelope parameters and determining appropriate modification ranges requires 2–3 h of effort from college students with entry level Energyplus skills, including consulting design guidelines, building codes, and prior simulation experience. By contrast, in the proposed agentic workflow, the LLM-assisted parameter schema construction completes within approximately 2–5 min, while embedding domain constraints through prompt-level parameter whitelisting.

Within this constrained search space, the agent identifies a compact set of envelope parameters with direct thermal performance implications. The selected parameters, their associated IDF object fields, baseline values, modification ranges, and expected physical effects are summarized in Table 5. Together, these parameters define a targeted design space that enables interpretable sensitivity analysis while avoiding combinatorial complexity.

Each parameter is then encoded into a structured JSON schema that preserves traceability between high-level optimization intent and low-level model manipulation. The schema records the IDF object type, target field, baseline value, modification range, and expected directional effect, enabling deterministic, rule-based IDF transformations in subsequent workflow stages. This machine-readable representation serves both as an executable artifact for automated model generation and as a transparent optimization plan that can be reviewed by human designers prior to simulation execution. The complete parameter schema is provided in **Appendix A2**.

3.3.4. Automated IDF modification and simulation execution

Following parameter schema construction, the workflow performs automated IDF modification and batch simulation through a rule-based execution stage. The structured JSON schema is deterministically translated into parametric IDF variants via category-specific operations, including building orientation adjustment, window area scaling using multiplier coefficients, and envelope material property modification. For the present case study, this process generates nine single-parameter variants, as documented in **Appendix A3**.

In conventional manual workflows, sequentially executing and monitoring simulations for the nine parametric models typically requires approximately 30 min of human effort, as each run must be manually launched, monitored for completion, and followed by manual initiation of the next simulation. By contrast, when orchestrated through the MCP simulation tool within the agentic workflow, the entire batch simulation stage is executed fully automatically and completes in approximately 10 min, substantially reducing human intervention and improving overall modelling efficiency.

Each simulation run produces a dedicated output directory containing annual energy summaries, hourly load profiles, and execution logs. A lightweight state registry maps each parametric variant to its corresponding output location, enabling systematic result retrieval and comparative analysis in subsequent workflow stages. The inputs and outputs of the simulation runner are summarized in List 1.

Table 5. Parameter schema from LLM-guided reasoning for automated IDF modification.

Parameter	IDF Object	Field	Baseline	Range	Physical Effect
Orientation	BUILDING	North_Axis	0°	0–180°	Redistribute solar gains
Window area	FENESTRATIONSURFACE	Multiplier	1.0	0.5–0.9	Reduce solar heat gain
Wall absorptance	MATERIAL	Solar Absorptance	0.6	0.2–0.5	Reflect solar radiation
Roof absorptance	MATERIAL	Thermal Absorptance	0.9	0.2–0.5	Reduce roof heat gain
Roof insulation	MATERIAL	Thickness	0.0869 m	0.1–0.2 m	Increase R-value

List 1. Simulation Runner Inputs & Outputs

Input = Baseline IDF and modified IDFs; MCP simulation_manager; Mounted model path and output directories; Container environment (EnergyPlus MCP server)

Output =

Baseline simulation folder (baseline_run_latest)

One simulation folder per variant

Process logs and timing information

Updated state.baseline_dir, state.sim_output_dirs

This part automates EnergyPlus simulations for baseline and modified IDF files, avoiding redundant baseline runs.

It uses EnergyPlusV25-1-0 to simulate

3.3.5. Sensitivity result analysis and visualization

Following the batch simulation execution, the workflow performs automated result parsing and sensitivity analysis to quantify the impact of individual envelope parameters on building thermal performance. A parsing agent systematically extracts key indicators from EnergyPlus CSV and ESO outputs, including annual heating and cooling loads and peak demand metrics. The baseline model establishes reference values of 1967.92 W for cooling load and 892.79 W for heating load, against which all parametric variants are evaluated.

Sensitivity scores are computed for each variant as normalized deviations from the baseline, capturing the relative influence of single-parameter perturbations on combined thermal demand. This single-parameter sensitivity strategy enables clear attribution of performance changes while avoiding the complexity of full factorial exploration. As summarized in Table 6, building orientation exhibits the highest quantitative sensitivity, with the 180° rotation variant yielding a sensitivity score of 51.99%, followed by the 90° orientation (18.1%), roof insulation thickness (7.0%), and window area reduction to 70% of baseline (3.6%).

Beyond numerical ranking, the workflow incorporates an LLM-based interpretation agent to assess whether highly sensitive parameters align with the user's stated optimization objective (*'reduce cooling load while maintaining comfort'*). While orientation changes dominate sensitivity rankings, the LLM correctly identifies that a 180° rotation, despite its high sensitivity, increases cooling demand in a cooling-dominated climate, rendering it counterproductive to the optimization goal. In contrast, envelope material modifications and moderate window

area reductions exhibit more consistent and goal-aligned performance trends.

The interpretation agent synthesizes quantitative sensitivity results with qualitative design reasoning to generate structured recommendations. For the present case study, the workflow prioritizes roof insulation enhancement and window area reduction to 70–80% of baseline as effective strategies, while cautioning that orientation changes require location-specific solar analysis. This interpretive layer transforms raw simulation outputs into actionable design insights suitable for early-stage decision-making by non-specialist users.

All sensitivity inputs and outputs are summarized in **List 2**, with complete ranked results provided in machine-readable CSV and JSON formats. The workflow further generates comparative charts and temporal load profiles through the MCP plot tool, ensuring that analysis and visualization remain fully reproducible within the same containerized execution environment.

List 2. Result Parsing & Sensitivity Inputs & Outputs

Input = All simulation result directories from Step 3; EnergyPlus output tables (.csv, .eso, etc.); Normalization formulas for heating/cooling loads; Sensitivity scoring algorithm

Output = Baseline totals → Cooling: 1967.92 W, Heating: 892.79 W

Among all modified simulations,

****Building_Orientation_180 deg ** showed the highest sensitivity with an overall change of **51.99%**.**

This indicates that the parameter has the strongest quantitative impact on the user's optimization goal. The complete ranked sensitivity list is available in the attached CSV/JSON files."}

3.3.6. Computational cost and reproducibility analysis

Based on experimental measurements across ten repeated runs, each workflow execution required approximately 9–12 min on average. For comparison, a graduate student with 1–2 years of BEM experience typically requires about 2.5 h to perform the same analysis, model modification, and simulation sequence manually. This indicates an order-of-magnitude reduction in human effort when using the automated workflow. To evaluate computational efficiency and account for LLM non-determinism, the complete workflow was executed 10 times under identical inputs. Token usage for every LLM call was recorded via API-reported usage statistics. Table 7 summarizes the total token consumption per run (mean $\pm$ standard deviation). The average token usage per run is 47,085 tokens, with a standard deviation of 7,912 tokens ($\sim 16.8\%$), reflecting variability in the number of analytical reasoning calls rather than instability in core model behaviour. GPT-5 is invoked once per run

Table 6. Top 5 sensitivity ranking results of Agent 3c

Modified parameter	Sensitivity score
Orientation_180°	52.0%
Orientation_90°	18.1%
Roof Insulation Thickness	7.0%
Window_to_Wall_Ratio_70%	3.6%
Window_to_Wall_Ratio_80%	2.6%

Table 7. LLM token consumption statistics across repeated workflow executions .

Metric	Value
Runs	10
Mean total tokens	47085.0
Standard deviation	7912.1
Min	34793
Max	63120

Table 8. Per-model token usage.

Model	Calls	Mean tokens per call	Std
GPT5	10	7,352.5	626.7
GPT4-mini	85	4,674.4	61.9

for schema reasoning and exhibits low variability (standard deviation of 627 tokens), while GPT-4o-mini handles repeated analytical tasks and contributes most of the run-to-run token fluctuation. Across 10 repeated runs, the top-ranked sensitive parameters remained consistent (orientation, roof insulation, window area), with only minor variations in LLM-generated explanatory phrasing. Despite stochastic differences in intermediate reasoning steps, the final optimization conclusions remained consistent across trials, demonstrating functional stability of the agentic workflow.

Additional variability arises from LLM-induced differences in workflow structure. During the schema construction, the LLM may select different numbers of envelope parameters across runs, resulting in varying numbers of IDF variants, simulation executions, and downstream analysis steps. Consequently, the number of GPT-4o-mini calls in the sensitivity analysis stage is not constant, leading to differences in total token consumption. This reflects agent-driven workflow branching rather than instability in individual model outputs. Despite these structural variations, the most influential parameters and overall optimization conclusions remained consistent. Table 8 further shows that GPT-5 was invoked once per run with low variability, while GPT-4o-mini accounts for most repeated analytical calls and the majority of run-to-run token variation.

4. Discussion

4.1. Toward a new BEM workflow paradigm

The integration of MCP with building energy modelling represents a fundamental shift in how practitioners interact with simulation tools. Traditional BEM workflows require users to navigate complex graphical interfaces or edit text-based input files directly, demanding specialized knowledge of object schemas, field definitions, and

inter-object dependencies. This expertise barrier has historically limited BEM adoption to specialists with years of training.

MCP-enabled workflows lower this barrier substantially. By exposing simulation capabilities through a standardized tool interface, MCP allows AI assistants to mediate between natural language intent and technical execution. Users express goals conversationally, and the AI assistant handles tool selection, input formatting, and result synthesis. The demonstrations in Section 3 show that tasks requiring 30 to 60 min of manual effort can be completed in under 10 min through conversational interaction.

Importantly, this paradigm augments rather than replaces professional judgment. The AI assistant provides comprehensive information, identifies potential issues, and generates visualizations, but the practitioner retains decision-making authority. Climate-appropriate retrofit selection, code compliance interpretation, and cost–benefit tradeoffs remain human responsibilities informed by AI-synthesized data. This collaborative model preserves professional accountability while dramatically improving workflow efficiency.

The educational implications are equally significant. Conversational BEM interactions make the modelling process transparent and explorable. Students and early-career professionals can learn by doing, asking questions about model behaviour and receiving immediate, contextualized responses. The AI assistant serves as both tool and tutor, explaining not just what the model contains but why certain configurations matter for energy performance.

4.2. Comparing integration paradigms

The two paradigms demonstrated in Section 3 offer complementary strengths suited to different use cases.

- **Conversational integration** places the user at the centre of an iterative dialogue. The practitioner steers exploration through natural language queries, with the AI assistant invoking MCP tools on demand. This approach excels for exploratory tasks where objectives may evolve based on intermediate findings. Model inspection, envelope analysis, and retrofit scenario development benefit from the flexibility to change direction as insights emerge. The conversational record also provides full transparency: every tool invocation is visible, creating an auditable trail of decisions.
- **Agentic workflow integration** coordinates multiple specialized agents that operate with minimal human intervention. After the user specifies a high-level

objective, the system autonomously decomposes the task, invokes appropriate tools, and iterates toward completion. This paradigm suits well-defined tasks with clear success criteria, such as sensitivity analysis across parameter ranges. The autonomous execution enables systematic coverage that would be tedious to conduct manually through repeated conversational exchanges.

Neither paradigm is universally superior. Conversational integration offers control and adaptability at the cost of sustained user engagement. Agentic workflows provide automation and scalability but require well-specified objectives and robust error handling. In practice, hybrid approaches may prove most effective: conversational exploration to understand a model and define objectives, followed by agentic execution for systematic analysis, with conversational review of results.

4.3. Limitations and considerations

While the demonstrations in Section 3 illustrate the potential of MCP-enabled workflows, this work represents an early exploration rather than a production-ready solution. Several limitations warrant acknowledgment, both specific to this study and inherent to the current state of LLM-based automation in technical domains:

- **Scope and generalizability.** The demonstrations in this paper employed a relatively simple residential model with three thermal zones and ideal loads. Complex models with hundreds of zones, detailed HVAC systems, or unusual configurations may pose different challenges that this study does not address. This model represents a real experimental facility and was selected to isolate workflow characteristics from model-specific complexity. Scaling to larger buildings would introduce challenges primarily at the LLM reasoning level: context window limits with hundreds of zones, dense inter-object dependencies in explicit HVAC loops, and longer simulation runtimes extending agentic workflow cycles. MCP's tool interface does not inherently constrain model complexity. Additionally, the demonstrated tasks covered inspection, modification, simulation, and sensitivity analysis, but many BEM workflows remain unexplored, including HVAC system design, daylighting analysis, advanced debugging, and multi-objective optimization. Systematic validation across the full BEM task spectrum represents essential future work.
- **LLM reliability, user interaction and system design.** The quality of MCP-enabled workflows depends on both the capabilities of the underlying LLM and the

user's ability to formulate effective queries. LLMs can produce incorrect tool invocations, misinterpret user intent, or generate plausible but inaccurate explanations. When many MCP tools are available, the model may select inappropriate tools or become confused by overlapping functionalities, and LLM non-determinism means identical prompts may yield different tool sequences across sessions. Similarly, vague or poorly structured user queries can lead to suboptimal tool selection and incomplete analyses, particularly in the conversational paradigm where the user steers each interaction turn. These challenges can be mitigated through careful system design, including tool consolidation, clear tool descriptions, validation checks, and human oversight for consequential decisions. The conversational paradigm's transparency, where all tool invocations and parameters are visible, provides an inherent error-detection mechanism that partially compensates for both LLM errors and imprecise user queries.

- **Infrastructure and cost.** Token consumption and API latency accumulate across extended sessions, making scripted approaches potentially more efficient for large-scale parametric studies. MCP's provider-agnostic design means the same EnergyPlus-MCP server works with any compatible client and LLM provider, so cost and configuration depend on practitioner choices rather than the protocol itself. Cost is primarily driven by token consumption, which varies with LLM quality: models with stronger tool-calling capabilities tend to require fewer interaction turns. Configuration difficulty reflects users' familiarity with their chosen environment (e.g. Claude Desktop, VS Code, GooseAI) rather than MCP-specific complexity. Additionally, effective AI-assisted BEM workflows benefit from access to domain knowledge sources such as building codes, material property databases, and benchmark repositories. Integrating these resources as MCP-accessible servers would enhance the AI assistant's ability to provide context-aware recommendations but requires infrastructure development beyond the simulation engine itself.

4.4. Future research directions

The demonstrations in this paper establish foundational workflows for MCP-enabled BEM, but significant opportunities remain to expand both scope and capability. We identify three directions that would advance the BEM field toward production-ready, end-to-end BEM automation.

- **Data-to-model automation.** Current workflows assume a pre-existing EnergyPlus model, but model creation itself remains labor-intensive. Future work should explore agentic systems that ingest heterogeneous data sources, including architectural drawings, basis of design documents, specifications, and BIM models, to generate simulation-ready inputs with minimal manual intervention. MCP servers exposing BIM platforms and document parsing tools could enable seamless data extraction and translation, substantially reducing the front-end burden of BEM projects.
- **Advanced agentic capabilities.** The sensitivity analysis demonstrated in Section 3.3 represents a relatively straightforward agentic task. More sophisticated applications include automated model calibration against measured data, multi-objective optimization balancing energy, cost, and comfort, and integration with digital twins for real-time monitoring and predictive simulation. These tasks require multi-agent architectures with dynamic coordination, persistent memory across sessions, and robust error recovery. Developing these capabilities would extend MCP-enabled workflows from design-phase analysis to operational building management.
- **Evaluation and trust.** Rigorous benchmarking with standardized test suites remains essential for comparing integration paradigms, LLM providers, and prompt strategies. Beyond performance metrics, future work should address validation frameworks that catch errors before they propagate, explainability mechanisms that trace recommendations to specific tool outputs, and privacy-preserving approaches for organizations with strict data governance. Understanding human-AI collaboration patterns, including how practitioners develop trust and identify errors, will shape the design of tools suitable for professional deployment. Comparative studies across LLM providers and user expertise levels would further clarify how model capability and query formulation skill influence workflow outcomes, informing both tool selection guidelines and user training strategies. Systematic scalability evaluation against progressively complex models, from commercial buildings with detailed HVAC to campus-scale models, would establish empirical boundaries for current LLM reasoning capacity and identify where hierarchical decomposition becomes necessary.

5. Conclusions

This paper introduced an MCP-enabled agentic framework for integrating AI assistants with building energy modelling workflows. Two complementary paradigms

were demonstrated: conversational integration for exploratory tasks such as model inspection, envelope analysis, and retrofit scenario development, and agentic workflow integration for systematic automation of sensitivity analysis. Both paradigms achieved substantial efficiency gains while maintaining full transparency through visible tool invocations. Importantly, the framework augments rather than replaces professional expertise, with AI handling tool orchestration, data synthesis, and visualization generation while practitioners retain authority over design decisions and cost–benefit tradeoffs. The introduced agentic AI BEM workflows not only extend the adoption of BEM to non-professional modellers but also significantly improve productivity of professional modellers.

Building on this foundation, future research can extend the approach to complex commercial building models and detailed HVAC system design, develop rigorous quantitative benchmarks for comparing integration paradigms, and explore data-to-model automatic BEM model creation from heterogeneous sources such as architectural drawings and BIM. Advanced agentic capabilities including model calibration against measured data and digital twin integration for real-time building operations represent promising directions.

Acknowledgements

To be added after the review.

Author contributions

CRedit: **Han Li:** Conceptualization, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **Liang Zhang:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Huiwen Zhou:** Formal analysis, Visualization, Writing – original draft, Writing – review & editing; **Tianzhen Hong:** Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by US Department of Energy: [Grant Number DE-AC02-05CH11231].

Data availability statement

The EnergyPlus-MCP is open source and available in Github at <https://github.com/LBNL-ETA/EnergyPlus-MCP>. The Appendix lists the scripts used in the paper. Other data (e.g. the residential energy model) will be provided upon request.

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Appendix

A1. Step 0: IDF inspection inputs & outputs

```

Input = Baseline IDF path provided by user; Mounted directory for container execution
MCP tools: model_preflight, envelope_manager, internal_load_manager, hvac_manager,
outputs_manager

Output =
=====

1 PREFLIGHT VALIDATION
=====

📁 Resolving paths...
Using Tool: model_preflight
{
  "idf": {
    "input": "/workspace/user-models/iUnit_Golden.idf",
    "resolved": "/workspace/user-models/iUnit_Golden.idf",
    "exists": true
  },
  "weather": {
    "input": null,
    "resolved": null,
    "exists": false,
    "error": null
  }
}
🔍 Checking readiness...
Using Tool: model_preflight
{
  "verdict": false,
  "idf": {
    "input": "/workspace/user-models/iUnit_Golden.idf",
    "resolved": "/workspace/user-models/iUnit_Golden.idf"
  },
  "weather": {
    "input": null,
    "resolved": null
  },
  "idd": {
    "path": "/app/software/EnergyPlusV25-1-0/Energy+.idd",
    "exists": true
  },
  "issues": [
    "Validation failed: Error validating IDF file: unsupported operand type(s) for +:
'Idf_MSequence' and 'Idf_MSequence'"
  ]
}

```

```

}
 Getting model info...
Using Tool: model_preflight
Model basics for /workspace/user-models/iUnit_Golden.idf:
{
  "Building": {
    "Name": "Building",
    "North Axis": 0.0,
    "Terrain": "Suburbs",
    "Loads Convergence Tolerance": 0.04,
    "Temperature Convergence Tolerance": 0.4,
    "Solar Distribution": "FullExterior",
    "Max Warmup Days": 25,
    "Min Warmup Days": 6
  },
  "Site:Location": {
    "Name": "Untitled (01-01:31-12) (01-01:31-12)",
    "Latitude": 39.74,
    "Longitude": -105.18,
    "Time Zone": -7.0,
    "Elevation": 1829.0
  },
  "SimulationControl": {
    "Do Zone Sizing Calculation": "Yes",
    "Do System Sizing Calculation": "Yes",
    "Do Plant Sizing Calculation": "No",
    "Run Simulation for Sizing Periods": "No",
    "Run Simulation for Weather File Run Periods": "Yes",
    "Do HVAC Sizing Simulation for Sizing Periods": "No",
    "Max Number of HVAC Sizing Simulation Passes": 1
  },
  "Version": {
    "Version Identifier": "25.1"
  }
}

```

2 ENVELOPE INSPECTION

```

 Inspecting envelope...
Using Tool: envelope_manager
 Summary:
- total_surfaces: 34
- exterior_surfaces: 22
- interior_surfaces: 12
- total_constructions: 13
- used_constructions: 5

```

```
- total_materials: 18
- orphaned_constructions: 8
- orphaned_materials: 1
- missing_references: 0
```

3 INTERNAL LOADS INSPECTION

```
👤 Inspecting internal loads...
Using Tool: internal_load_manager
```

4 HVAC SYSTEMS INSPECTION

```
🔧 Discovering HVAC loops...
Using Tool: hvac_manager
```

```
📊 Summary:
- total_plant_loops: 0
- total_condenser_loops: 0
- total_air_loops: 0
- total_zones: 3
```

5 OUTPUT VARIABLES/METERS LISTING

```
📈 Listing outputs...
Using Tool: outputs_manager
```

A2. Parameter schema generation inputs & outputs

Input = Baseline model info from Step 0; IDF inspection results;
User Goal: "Reduce cooling load while maintaining comfort"

```

Output =
{
  "goal": "reduce cooling load while maintaining comfort",
  "parameters_to_modify": {
    "Building_Orientation": {
      "idf_object": "BUILDING",
      "category": "Building",
      "object_name": "Unnamed_0",
      "field": "North_Axis",
      "original_value": 0.0,
      "expected_effect": "adjust",
      "suggested_range": [
        0,
        180
      ],
      "unit": "deg"
    },
    "Window_to_Wall_Ratio": {
      "idf_object": "FENESTRATIONSURFACE:DETAILED",
      "category": "Window",
      "object_name": "Unnamed_1",
      "field": "Multiplier",
      "original_value": 1.0,
      "expected_effect": "decrease",
      "suggested_range": [
        0.5,
        0.9
      ],
      "unit": ""
    },
    "Exterior_Wall_Solar_Absorptance": {
      "idf_object": "MATERIAL",
      "category": "Material",
      "object_name": "Unnamed_0",
      "field": "Solar_Absorptance",
      "original_value": 0.6,
      "expected_effect": "decrease",
      "suggested_range": [
        0.2,
        0.5
      ],
      "unit": ""
    },
    "Roof_Thermal_Absorptance": {
      "idf_object": "MATERIAL",
      "category": "Material",
      "object_name": "Unnamed_2",
      "field": "Thermal_Absorptance",
      "original_value": 0.9,
      "expected_effect": "decrease",
      "suggested_range": [
        0.2,
        0.5
      ],
      "unit": ""
    },
    "Roof_Insulation_Thickness": {
      "idf_object": "MATERIAL",
      "category": "Material",
      "object_name": "Unnamed_0",
      "field": "Thickness",
      "original_value": 0.0869,
      "expected_effect": "increase",
      "suggested_range": [
        0.1,
        0.2
      ],
      "unit": "m"
    }
  },
  "total_parameters": 5
}

```

A3. Parameter modification inputs & outputs

Input = Parameter schema from Step 1; Rule-based modifier logic (window scaling, geometry transforms, orientation adjustments, etc.); Baseline IDF file

Output = Found 5 parameters to modify using SMART RULES.

Processing: Building_Orientation

Category: Building | Field: North_Axis

Type: ORIENTATION | Variants: 3

[1] 0° → 0° (Baseline)

[2] 0° → 90° (Rotation)

[3] 0° → 180° (Rotation)

Generated 3 IDF files

Processing: Window_to_Wall_Ratio

Category: Window | Field: Multiplier

Type: WINDOW_GEOMETRY | Variants: 3

[1] Scale to 70% (WWR ↓51%)

[2] Scale to 80% (WWR ↓36%)

[3] Scale to 90% (WWR ↓19%)

Generated 3 IDF files

Processing: Roof_Material_Conductivity

Category: Material | Field: Conductivity

[1] 0.04 → 0.20 (+400%) | Midpoint of [0.1-0.3]

Generated 1 IDF file

Processing: Exterior_Wall_Solar_Absorptance

Category: Material | Field: Solar_Absorptance

[1] 0.6 → 0.35 (-41.7%) | Midpoint of [0.2-0.5]

Generated 1 IDF file

Total files created: 9

Change log saved: outputs/modified_idfs/modification_summary.txt